Linear Discriminant Analysis

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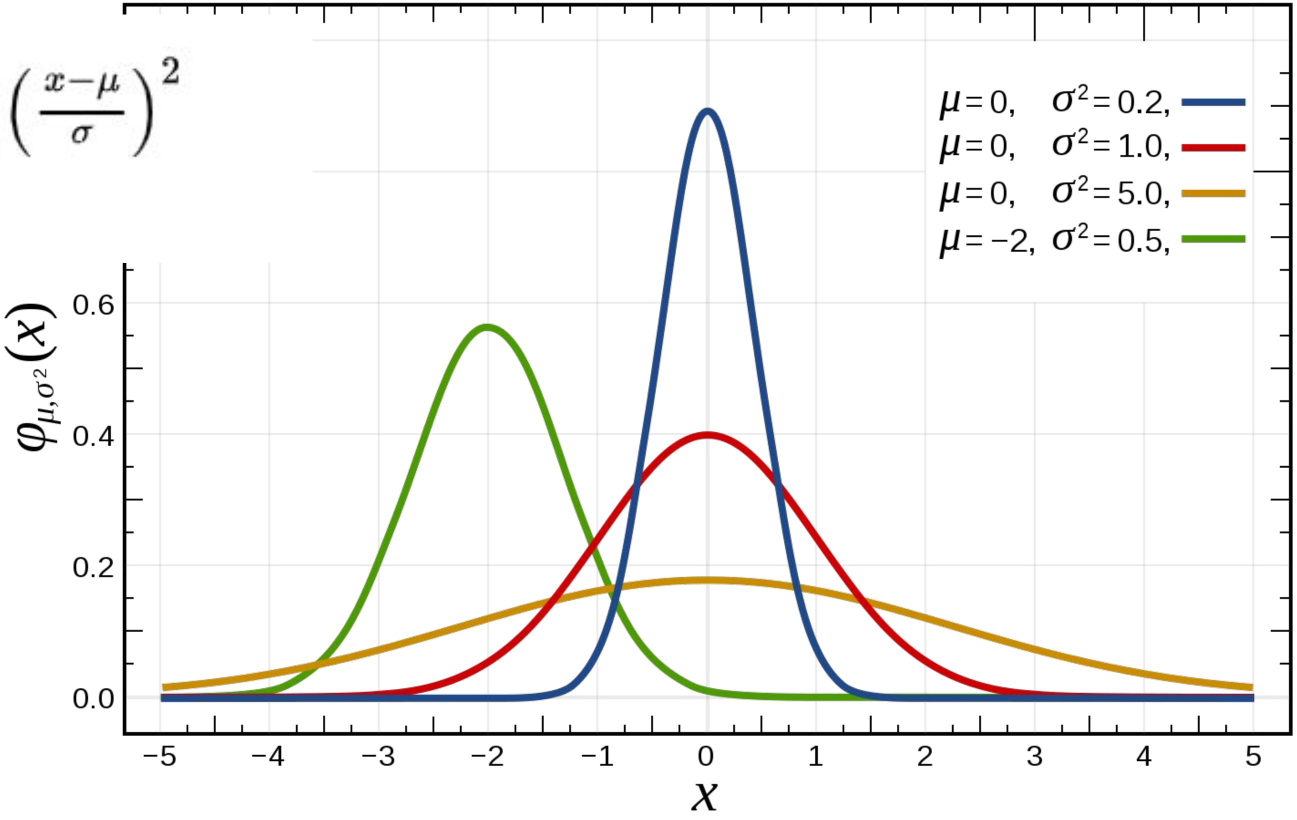
LDA

- Linear Discriminant Analysis (LDA) models each category (class) as a multivariate Gaussian distribution with equal variances and uses Bayes' Theorem to estimate probabilities.
- Fischer's linear discriminant is a variety of LDA
    - uses projection to reduce dimensionality
    - aids in separating categories

Normal Distribution

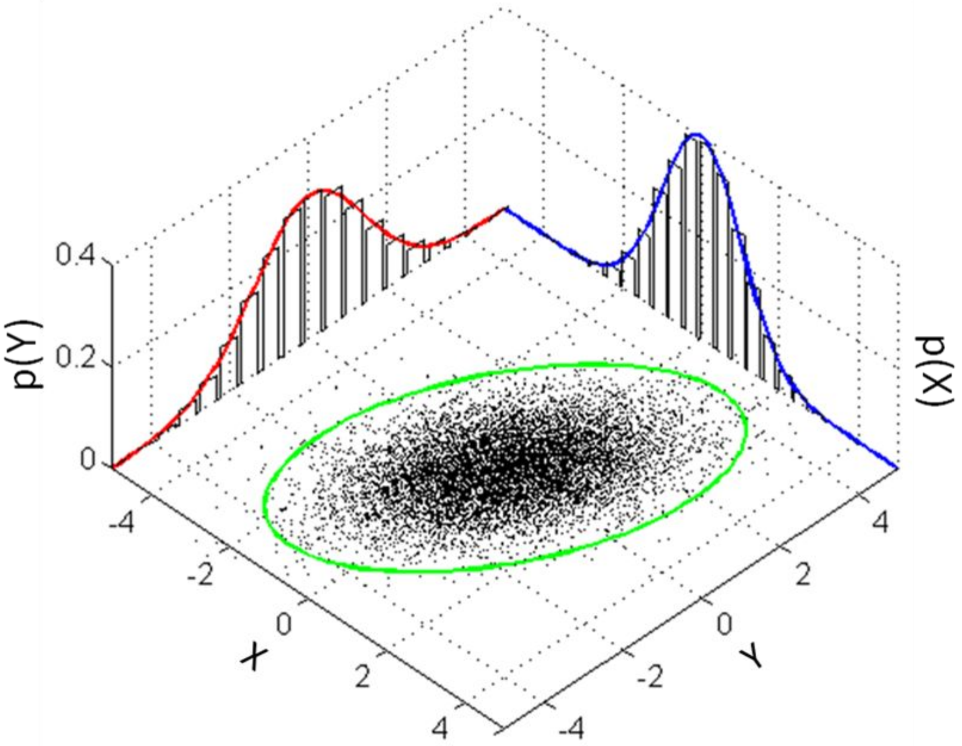
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



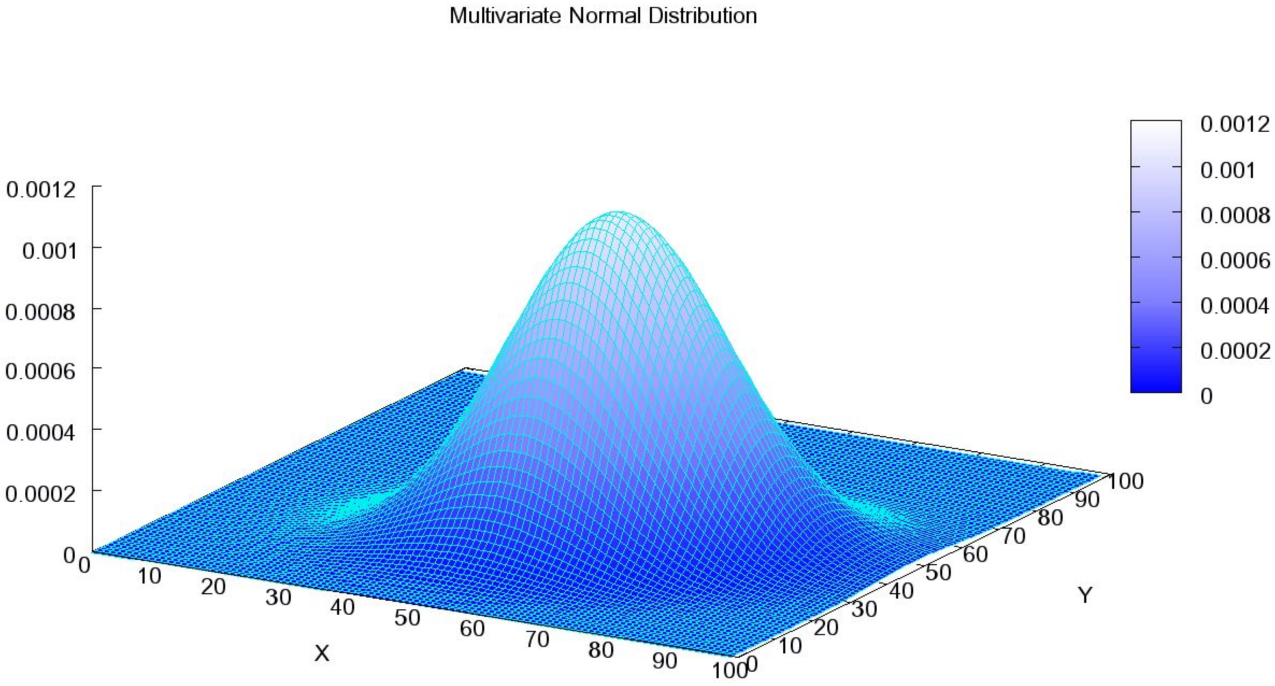
Multivariate Normal Distribution

- Generalization of the one-dimensional (univariate) normal distribution to higher dimensions.

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix},$$

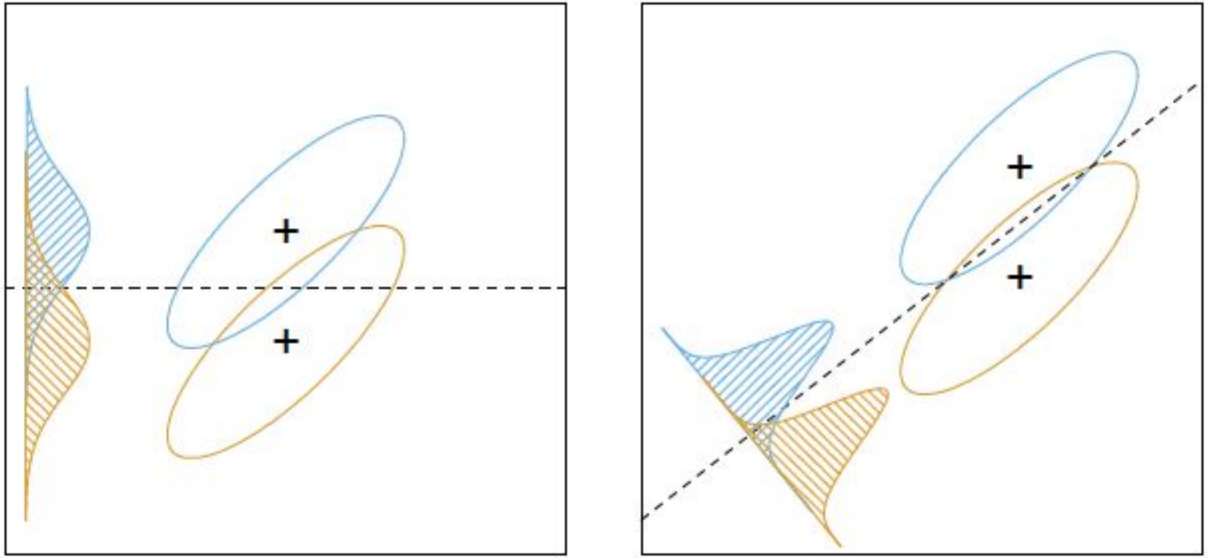


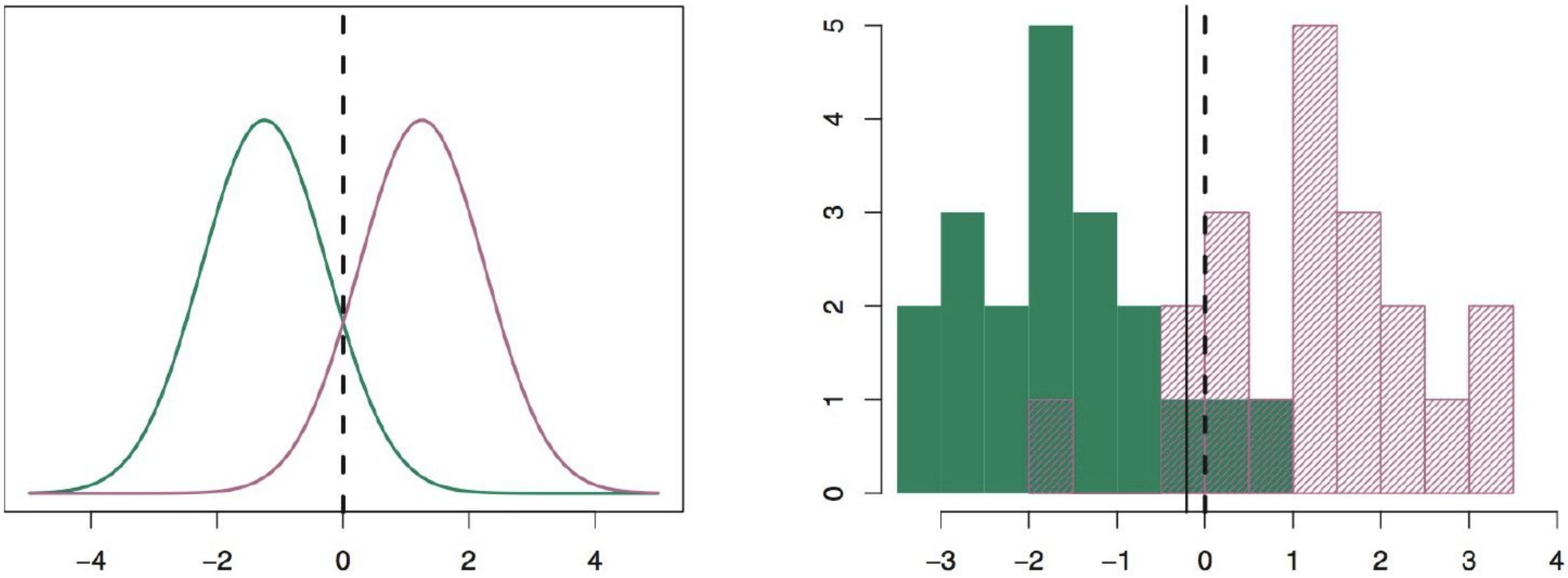
Multivariate Normal Distribution



Multivariate Normal Distribution

$$N(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$





−4
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Fish Packing Conveyor Belt

https://en.wikipedia.org/wiki/Salmon

https://en.wikipedia.org/wiki/Striped_bass

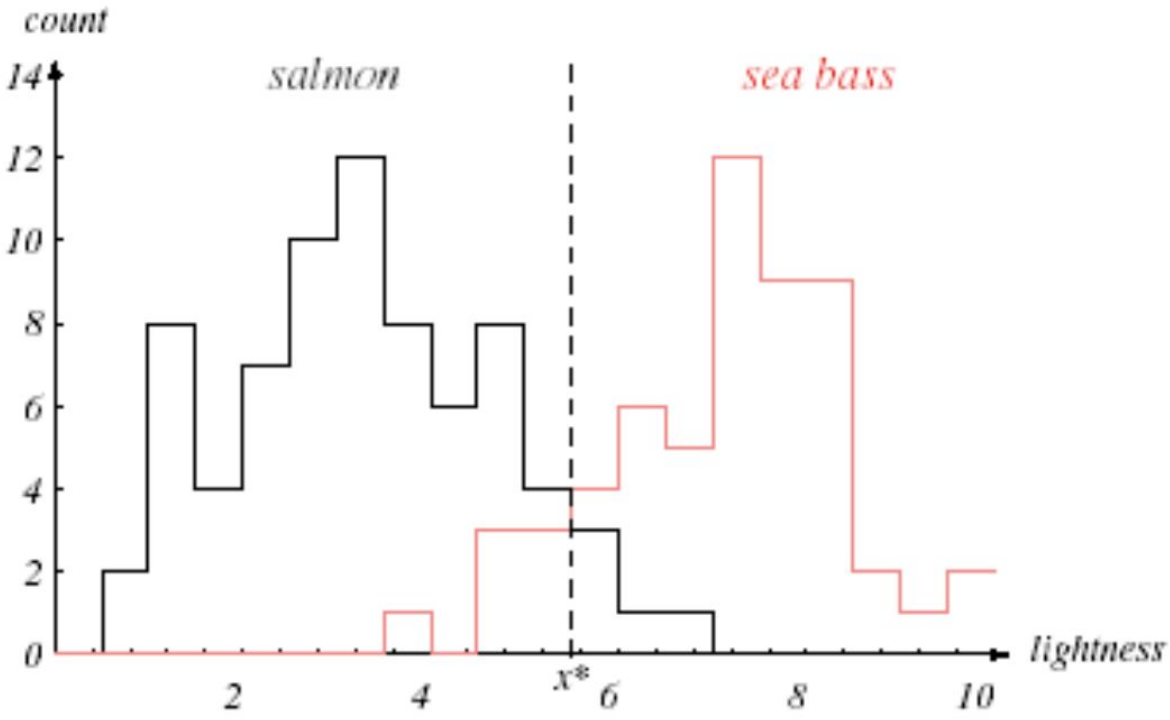


FIGURE 1.3. Histograms for the lightness feature for the two categories. No single threshold value x^* (decision boundary) will serve to unambiguously discriminate between the two categories; using lightness alone, we will have some errors. The value x^* marked will lead to the smallest number of errors, on average. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Two variables and Three classes

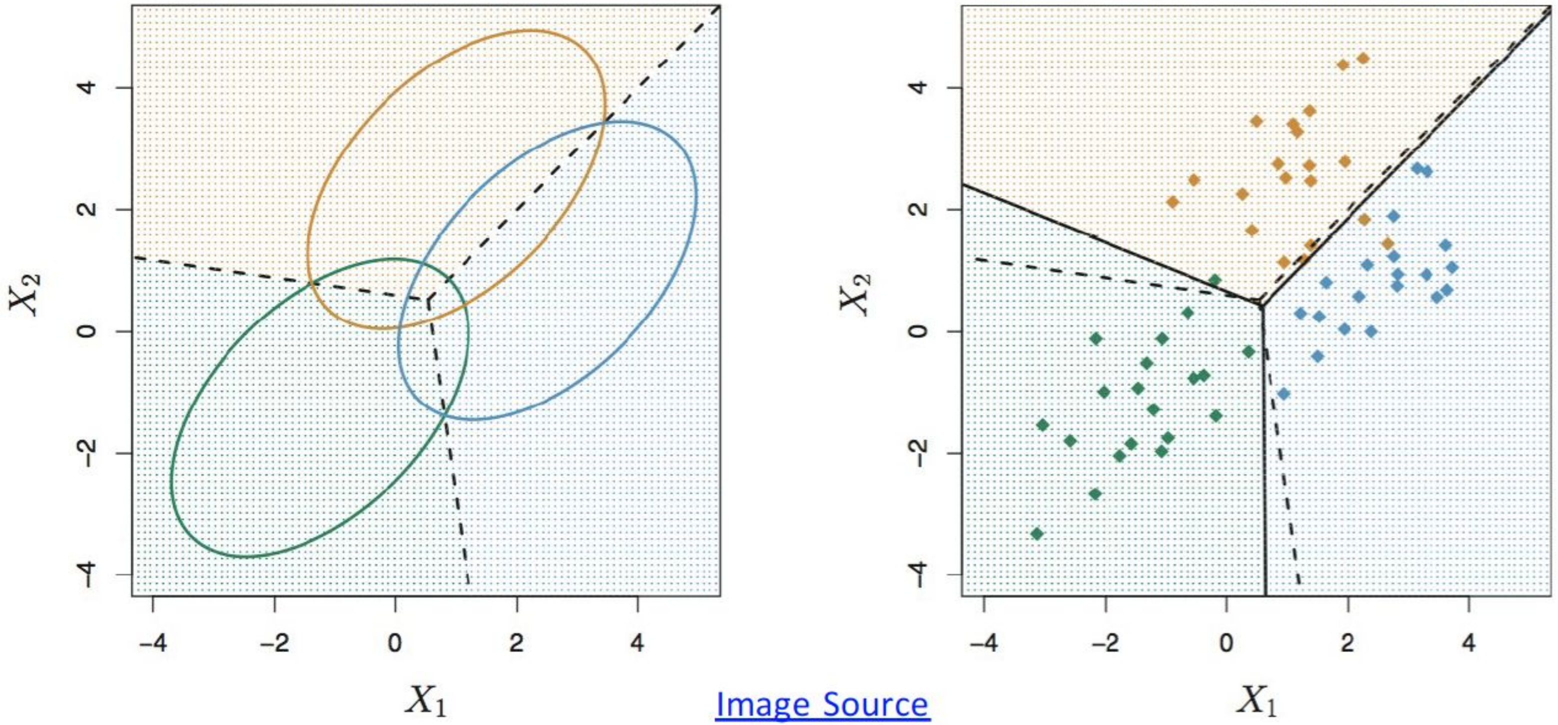
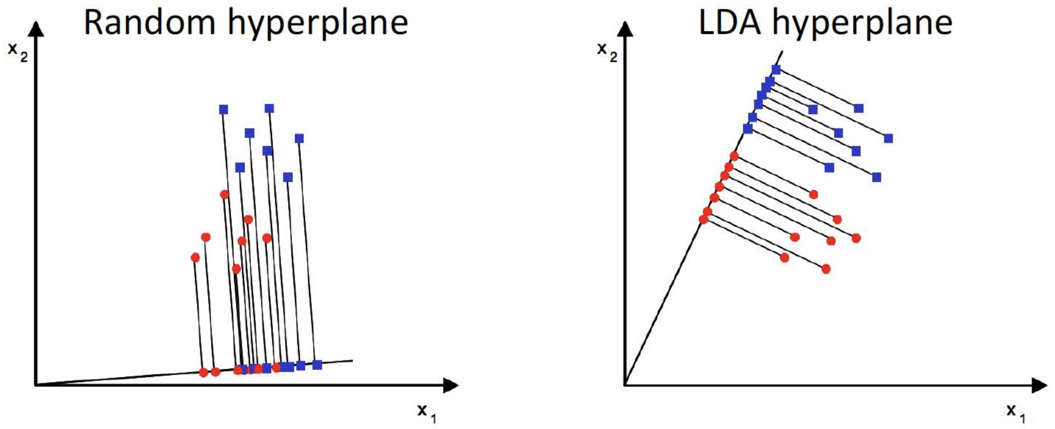


Image Source

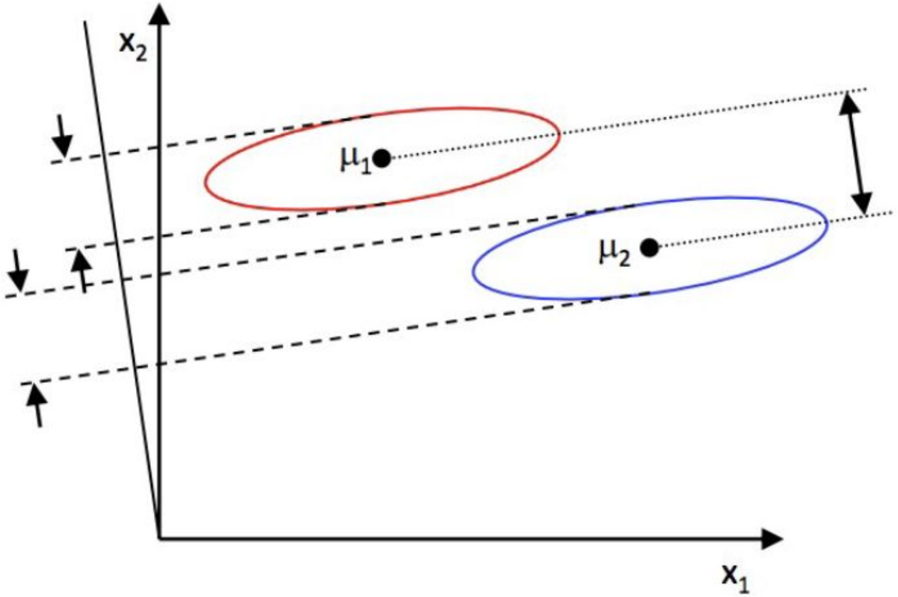
Fisher’s Linear Discriminant Analysis

- Fisher’s Linear Discriminant is a variety of LDA
- LDA is a generalization of FDA
- It uses projections similar to PCA to aid in classification.
- The data is projected to an optimal hyperplane that aims to simultaneously maximize between-class variance and minimize within-class variance.



Projection to direction W

$$y = \overrightarrow{w}^T \vec{x}$$



Means of the projected points

$$\vec{\mu}_i = \frac{1}{N_i} \sum_{\vec{x} \in \text{class } i} \vec{x}$$

$$\tilde{\mu}_i = \frac{1}{N_i} \sum_{x \in \text{class } i} \vec{w}^T \vec{x} = \vec{w}^T \vec{\mu}_i$$

$$\tilde{s}_i^2 = \sum_{y \in \text{class } i} (y - \tilde{\mu}_i)^2$$

Fisher’s LDA

maximize the distance between the projected means and minimize the projected within-class scatter.

$$J(\vec{w}) = \frac{|\tilde{\mu}_1 - \tilde{\mu}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

Within class and between class scatter

$$\overleftrightarrow{S}_B = (\vec{\mu}_2 - \vec{\mu}_1)(\vec{\mu}_2 - \vec{\mu}_1)^T$$

$$\overleftrightarrow{S}_w = \sum_{i \in \text{class } 1} (\vec{x}_i - \vec{\mu}_1)(\vec{x}_i - \vec{\mu}_1)^T + \sum_{i \in \text{class } 2} (\vec{x}_i - \vec{\mu}_2)(\vec{x}_i - \vec{\mu}_2)^T$$

$$
\begin{aligned}
\hat{s}_k^2 &= \sum_{i \in C_k} (y^{(i)} - \hat{u}_k)^2 \\
&= \sum_{i \in C_k} (\theta^T x^{(i)} - \theta^T u_k)^2 \\
&= \sum_{i \in C_k} \theta^T (x^{(i)} - u_k)(x^{(i)} - u_k)^T \theta \\
&= \theta^T S_k \theta
\end{aligned}
$$

$$
\begin{aligned}
\hat{s}_1^2 + \hat{s}_2^2 &= \theta^T S_1 \theta + \theta^T S_2 \theta \\
&= \theta^T S_W \theta
\end{aligned}
$$

$$
\begin{aligned}
(\hat{u}_2 - \hat{u}_1)^2 &= (\theta^T u_2 - \theta^T u_1)^2 \\
&= \theta^T (u_2 - u_1)(u_2 - u_1)^T \theta \\
&= \theta^T S_B \theta
\end{aligned}
$$

Rewriting the Maximization Criteria

$$J(\overrightarrow{w}) = \frac{\overrightarrow{w}^T \overleftrightarrow{S}_B \overrightarrow{w}}{\overrightarrow{w}^T \overleftrightarrow{S}_W \overrightarrow{w}}$$

$$
\begin{aligned}
\frac{\partial J(\theta)}{\partial \theta} &= \frac{\partial}{\partial \theta}(\frac{\theta^T S_B \theta}{\theta^T S_W \theta}) \\
&= (\theta^T S_W \theta)\frac{\partial(\theta^T S_B \theta)}{\partial \theta} - (\theta^T S_B \theta)\frac{\partial(\theta^T S_W \theta)}{\partial \theta} = 0 \\
\implies &= (\theta^T S_W \theta) 2 S_B \theta - (\theta^T S_B \theta) 2 S_W \theta = 0
\end{aligned}
$$

Divided by $\theta^T S_W \theta$:

$$
\begin{aligned}
&\implies (\frac{\theta^T S_W \theta}{\theta^T S_W \theta}) S_B \theta - (\frac{\theta^T S_B \theta}{\theta^T S_W \theta}) S_W \theta = 0 \\
&\implies S_B \theta - J S_W \theta = 0 \\
&\implies S_W^{-1} S_B \theta - J\theta = 0 \\
&\implies J\theta = S_W^{-1} S_B \theta \\
&\implies J\theta = S_W^{-1}(u_2 - u_1)(u_2 - u_1)^T \theta \\
&\implies J\theta = S_W^{-1}(u_2 - u_1)(\underbrace{(u_2 - u_1)^T \theta}_{c \in \mathbb{R}}) \\
&\implies J\theta = c S_W^{-1}(u_2 - u_1) \\
&\implies \theta = \frac{c}{J} S_W^{-1}(u_2 - u_1)
\end{aligned}
$$

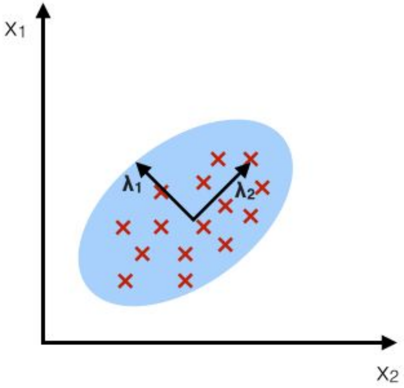
$$\vec{w}^* = \overleftrightarrow{S}_W^{-1}(\vec{\mu}_2 - \vec{\mu}_1).$$

Dimensionality Reduction

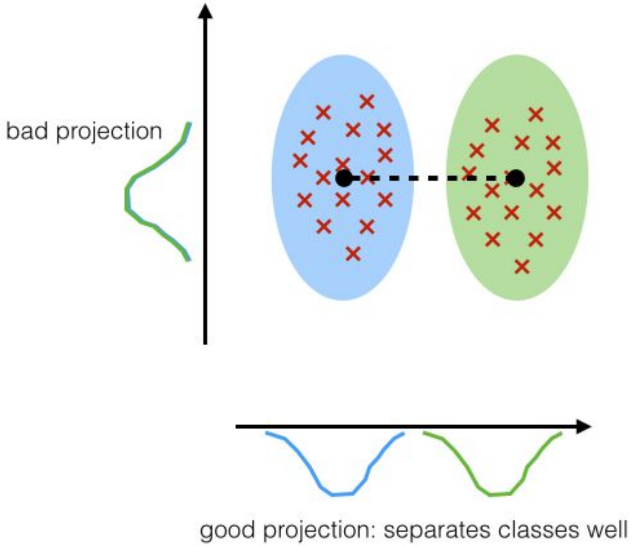
- If k classes, then at most k-1 feature projections
- If we have 100 variables but 10 classes then we will get 9 dimensions (huge reduction in dimensionality)
- But there is the danger of double dipping (it uses the labels)

LDA vs PCA

PCA:
component axes that maximize the variance



LDA:
maximizing the component axes for class-separation



https://sebastianraschka.com/Articles/2014_python_lda.html

Multi-class problem

Instead of one projection, we now need C − 1 (C is number of the classes)

$$y_i = \theta_i^T X \implies y = \Theta^T X$$

The optimal projection matrix Θ∗ is the one whose columns are the eigenvectors corresponding to the $S_W^{-1} S_B$

$$J(W) = \frac{|\hat{S}_B|}{|\hat{S}_W|} = \frac{|\Theta^T S_B \Theta|}{|\Theta^T S_W \Theta|}$$

$$\Theta^* = [\theta_1^* | \theta_2^* | \ldots | \theta_{C-1}^*]$$

LDA Five Steps

1. Compute the d-dimensional mean vectors for the different classes from the dataset.
2. Compute the scatter matrices (in-between-class and within-class scatter matrix).
3. Compute the eigenvectors (e1,e2,...,ed) and corresponding eigenvalues (λ1,λ2,..., λd) for the scatter matrices $S_W^{-1} S_B$
4. Sort the eigenvectors by decreasing eigenvalues and choose k eigenvectors with the largest eigenvalues to form a d×k dimensional matrix W (where every column represents an eigenvector).
5. Use this d×k eigenvector matrix to transform the samples onto the new subspace. This can be summarized by the matrix multiplication: Y=X×W (where X is a n×d-dimensional matrix representing the n samples, and y are the transformed n×k-dimensional samples in the new subspace).

LDA for classification

- Until now, we only reduced the dimension of the data points
- The projected data to the new space can subsequently be used to construct a classifier by using Bayes' theorem as follows

Bayes Theorem

$$P(A|B) = \frac{P(A \wedge B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Baye's theorem for LDA

$$\Pr(Y=k|X=x)=\frac{\Pr(X=x|Y=k)\Pr(Y=k)}{\sum_{i=1}^{K}\Pr(X=x|Y=i)\Pr(Y=i)}=\frac{f_k(x)\pi_k}{\sum_{i=1}^{K}\pi_i f_i(x)},$$

where

$$f_k(x)=\Pr(Y=k|X=x)$$

and

$$\pi_k=\Pr(Y=k)$$

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2{\sigma_k}^2}(x - \mu_k)^2\right).$$

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{i=1}^{K} \pi i \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_i)^2\right)}$$

Estimates for the quantities

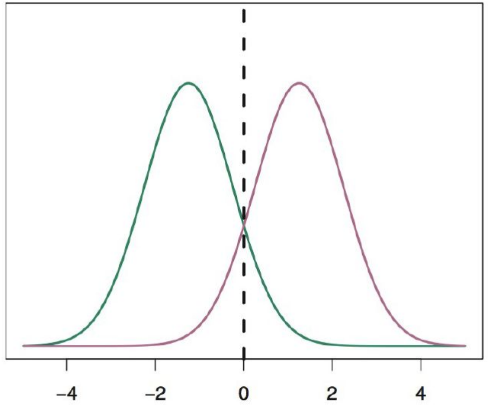
We calculate these for the projected input data in the new LDA space

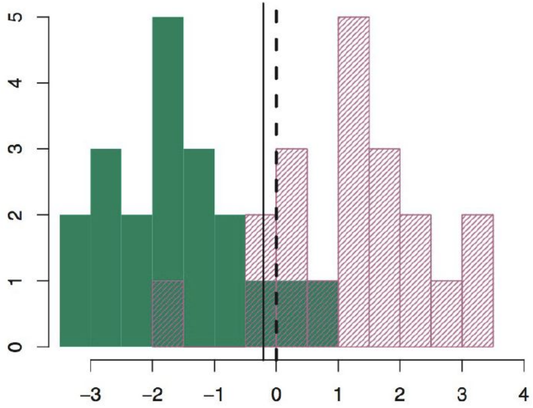
$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

$$\hat{\pi}_k = n_k/n$$

Two classes and single variable scenario





$$x = \frac{\mu_1^2 - \mu_2^2}{2(\mu_1 - \mu_2)} = \frac{\mu_1 + \mu_2}{2}$$

Multivariate Version

- We use a a multivariate Gaussian distribution, the rest is the same

$$f(\vec{x}) = \frac{1}{2\pi^{p/2}|\overleftrightarrow{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\vec{x} - \vec{\mu})^T \overleftrightarrow{\Sigma}^{-1}(\vec{x} - \vec{\mu})\right)$$

- Taking the logarithm and doing some algebra, we can write a Linear score function as:

$$\delta_k(x) = x^T \Sigma^{-1} m_k - \frac{1}{2} m_k{}^T \Sigma^{-1} m_k + \log pi_k \quad \ldots\ldots\ldots$$

We classify a sample to the class that has the highest Linear Score function for it